

Biology Practice Test 01

Format: ISC Subjective

Portion: Set 1 Midterm (Ch 2, 3, 4, 5, 6, 7, 8, 10, 13)

Time Allowed: 45 Minutes

Maximum Marks: 30

General Instructions:

- All questions are compulsory. Internal choice is provided in Section D.
 - Maintain neat handwriting and use a pencil for diagrams.
 - Stick strictly to the word limits and use ISC-specific terminology.
-

SECTION A (5 Marks)

Answer all questions. Each question carries 1 mark.

Question 1: Which of the following cells provide nutrition to the developing sperms? (a) Leydig cells (b) Sertoli cells (c) Spermatogonia (d) Primary spermatocytes

Question 2: Define *apomixis*.

Question 3: Sickle-cell anemia is an example of: (a) Point mutation (b) Frameshift mutation (c) Chromosomal aberration (d) Aneuploidy

Question 4: Name the enzyme that catalyzes the process of reverse transcription.

Question 5: Which of the following factors does *not* disturb the Hardy-Weinberg equilibrium? (a) Genetic drift (b) Random mating (c) Mutation (d) Gene flow

SECTION B (8 Marks)

Answer all questions. Each question carries 2 marks.

Question 6: Differentiate between Active Immunity and Passive Immunity with one example each.

Question 7: Explain the fundamental difference between the primary and secondary treatment of sewage.

Question 8: What are homologous and analogous organs? Give one example of an analogous organ.

Question 9: State any two essential features that an ideal contraceptive must possess.

SECTION C (12 Marks)

Answer all questions. Each question carries 3 marks.

Question 10: Briefly describe the Meselson-Stahl experiment and state its final conclusion regarding DNA replication.

Question 11: State Mendel's Law of Independent Assortment. Illustrate it using the phenotypic ratio of a standard dihybrid cross.

Question 12: Draw a neat, properly labeled diagram of a mature angiosperm embryo sac (highlighting polarities/ends clearly).

Question 13: Explain the following population interactions with one appropriate example for each: (a) Mutualism (b) Commensalism (c) Parasitism

SECTION D (5 Marks)

Answer any ONE of the following questions.

Question 14: (Option A) (a) With the help of a labeled diagram, explain the mechanism of the *lac* operon in *E. coli* in the presence of an inducer (lactose). (3 Marks) (b) Why is the regulation of the *lac* operon by a repressor referred to as "negative regulation"? (2 Marks)

OR

(Option B) (a) Describe the process of spermatogenesis in human males with a schematic representation. (3 Marks) (b) Mention the specific roles of Luteinizing Hormone (LH) and Follicle Stimulating Hormone (FSH) in this process. (2 Marks)